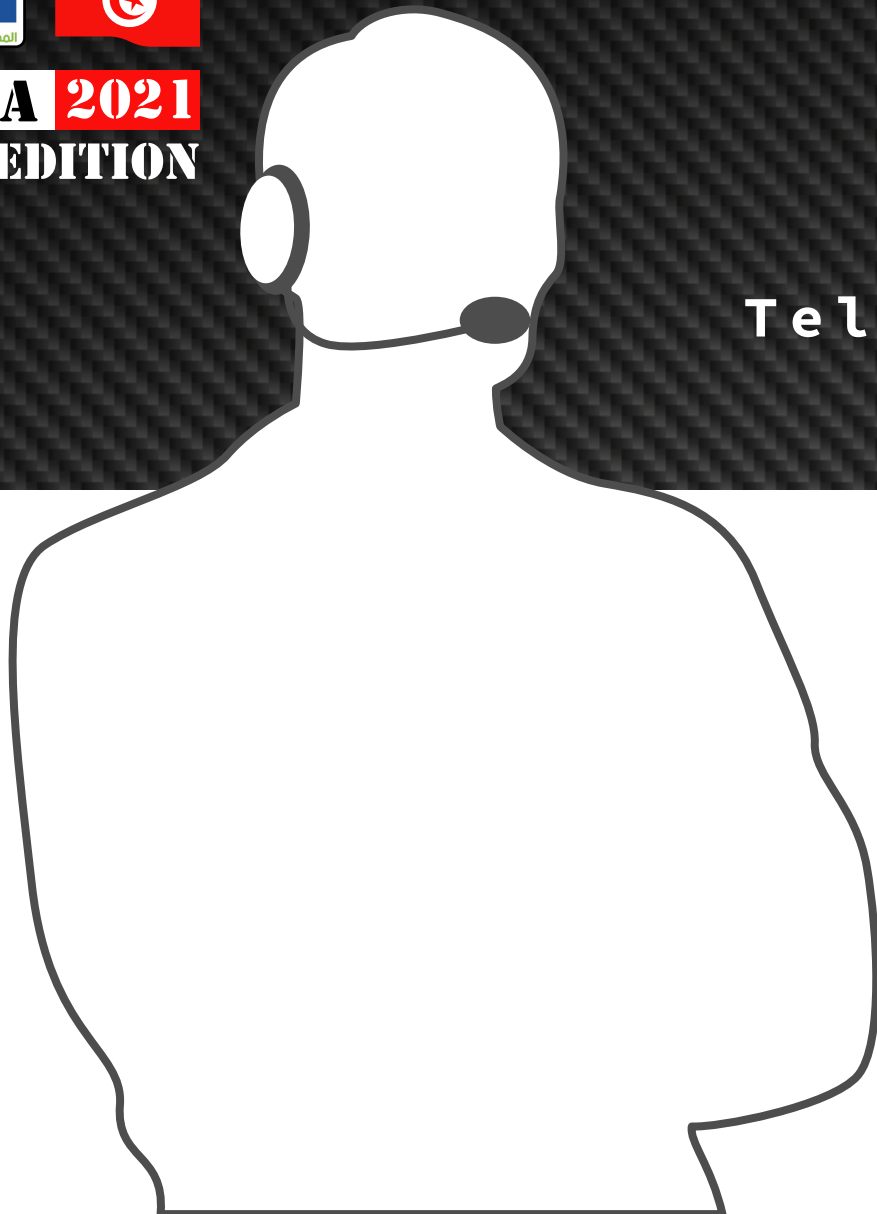




ENIMIA 2021
BLACK EDITION



Telemetry DATA File - Enim Team - Tunisia



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N e c e s s a r y D A T A

In addition to the position, speed and energy consumption data available on the existing telemetry system, the vehicle environment, temperature of components, ground inclination, tire pressure and data of critical components in the vehicle are also required. In addition, the current data transmission technology does not meet our needs in terms of transmission speed, so the data we need are:

Real Time Speed:

Vehicle's speed is a principle and direct factor of energy consumption. To measure the real time speed, we are going to use a magnetic speed sensor which measures the wheel's rotation rate "RPM", a simple multiplication with wheel's perimeter gives us the current linear speed.

Force applied on each wheel:

To ensure the stability of our vehicle, we need to maintain a well distributed charge between the three wheels. To do so, we are going to use a force gauge in each wheel.

This data will be used by the circuit engineer so he can advise the driver on how he should position himself in the car during the race. After that, the data will be stored and used in the future in order to develop better stability solutions to be used in future competitions.

Tire Pressure:

Having the correct tire pressure is extremely important for getting good energy mileage and the most life out of our tires. Our car has a recommended tire pressure, that will give the best contact surface that makes just the requested adherence in the circuit and reduces energy losses caused by wheels friction with the ground. A Manometer will be equipped so it shows and saves real time pressure.

Motor's Temperature:

The chief enemy of an electric motor's insulation system is heat, so we want to make sure the motor stays within limits for long winding life. To protect the marathon's drivers, prevent emergencies and study our motor's efficiency, we are going to use a PTC Thermistor Sensor that shows and saves real time motor's heat.

Calculator's Temperature:

Overheating causes damage to the circuit components and can cause fire, explosion, and injury. The use of a temperature sensor will provide us with information about our calculator real time environment's temperature in order to determine our calculator's status during the run.

Wind pressure (Drag):

To study the aerodynamics of our vehicle, knowing the wind pressure applied on the external surface is extremely important, that's why we're willing to use a "Soft sensor for measuring wind pressure".

Real Time Position:

A GPS Tracking unit will be installed in our vehicle, so the team can follow the real time car's position. As a result, we will be able to tell the driver about his position in the track, so he can reduce his speed in hard turns, or increase it in straight lines. Furthermore, an algorithm will be connected to the GPS tracker which increment the lap number each time our vehicle passes the finish line so that we can make the exact number of laps. Otherwise, the GPS tracking unit will help us in a major way in order to draw a proper cartography so that we can enhance the race line to take for the next essays.

N e c e s s a r y D A T A

Energy consumption:

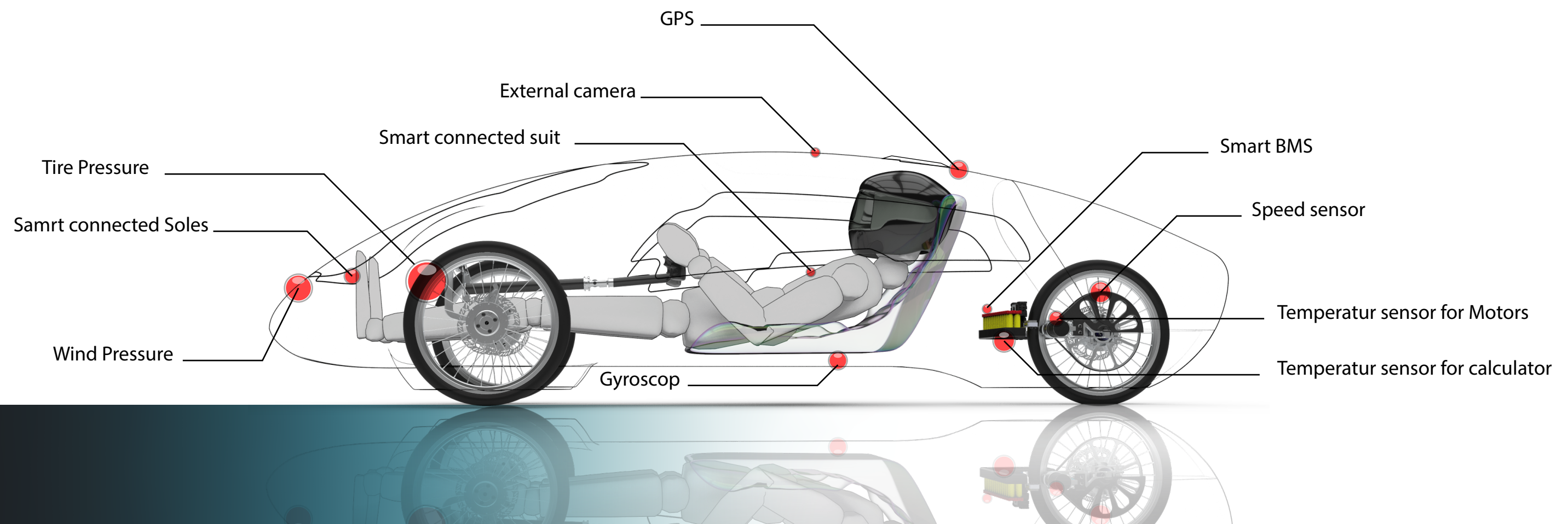
Energy consumption measurement tool (Joulemeter) for prototype concept provided by Shell organisers does not allow us to follow and save real time energy consumed. As we need this information to study our consumption parkour, we planned to use our own Joulemeter connected to the telemetry SMART CONNECTED BMS system to help reach our predefined goals.

Video and Audio

An audio and video communication between the pilot and the team is necessary, the team must have access to 3 video signals, one outside, another in the cockpit and one in the engine compartment. The latter is a thermal camera that the technical team will need to check and then predict possible problems.

Pilot's state of health

The team will have access to the health status of the pilot such as temperature, heart rate, humidity and percentage of oxygen through the connected suit of the pilot with the medical sensors. The goal is to analyze the driver's handling during the race and evaluate his fatigue and stress in relation to different driving situations.



U s e o f D A T A

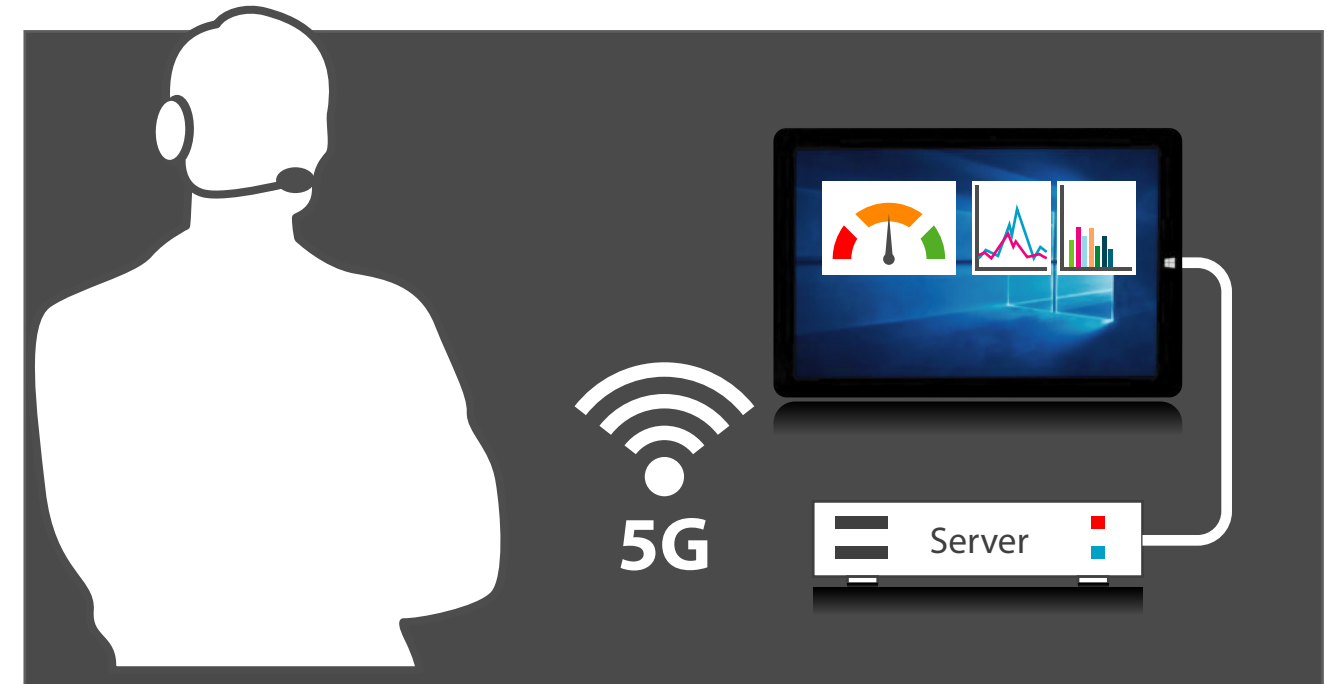
The team will use the data in 2 modes:

Real time during the race

The data transmitted from the vehicle passes through real-time computation loops in a server located next to the team. These computation loops run via an artificial intelligence based on an algorithm developed and parameterized according to the physical specifications of the vehicle and generate instructions to be applied by the driver to minimize energy consumption.

For example, the loop calculating of the distance to be crossed by the prototype with its own inertia gives the instantaneous speed, the angle of the slope and the GPS position of the vehicle in relation to the mapping of the circuit traced in the previous laps, an indication of the ideal speed that the driver must reach by acting on the gas pedal.

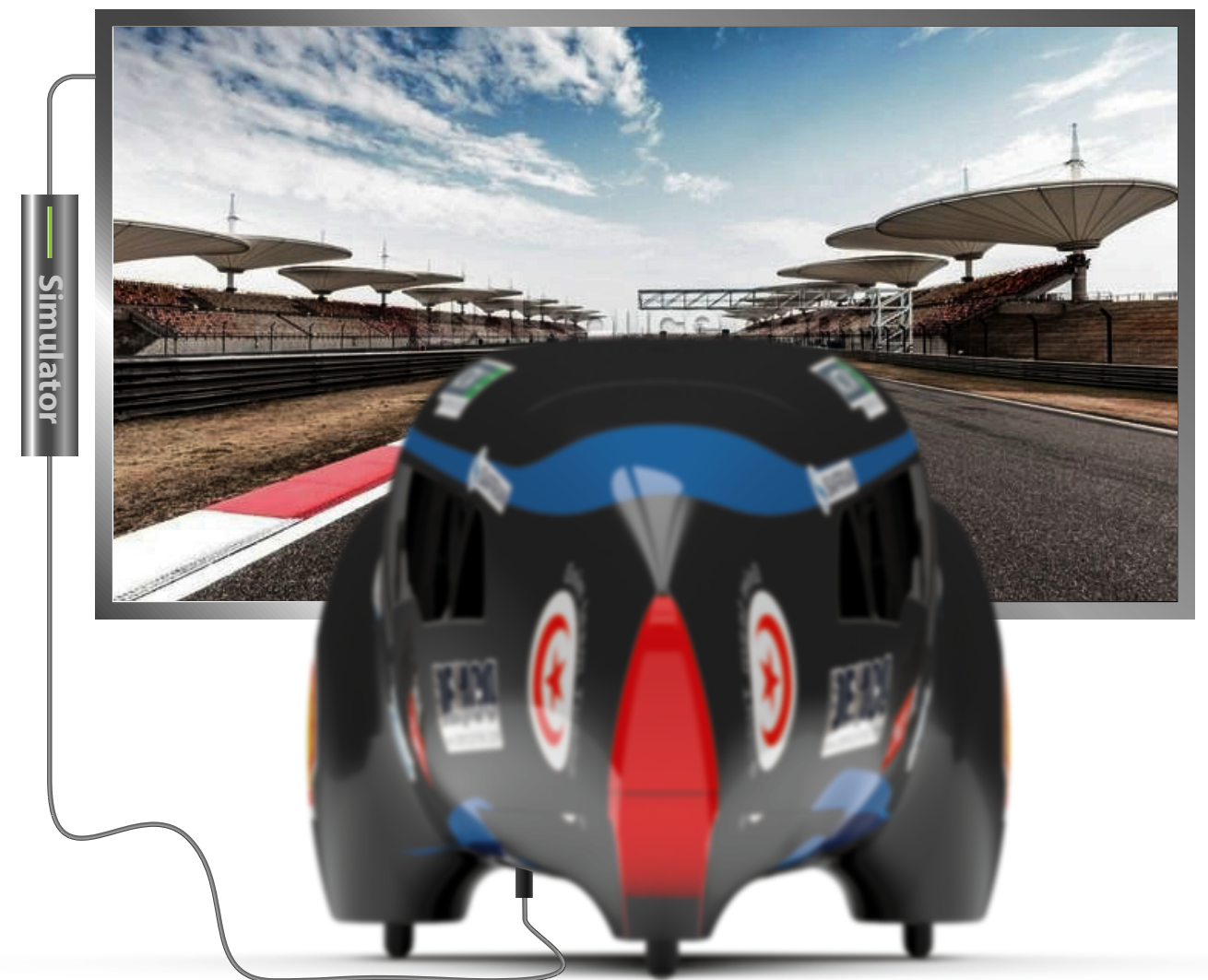
The driver will therefore not need to brake when cornering because the speed at this level is already closely estimated by calculation to reduce the excess energy consumed at an acceleration dissipated unnecessarily by braking.



U s e o f D A T A

Statistics and simulations after the race

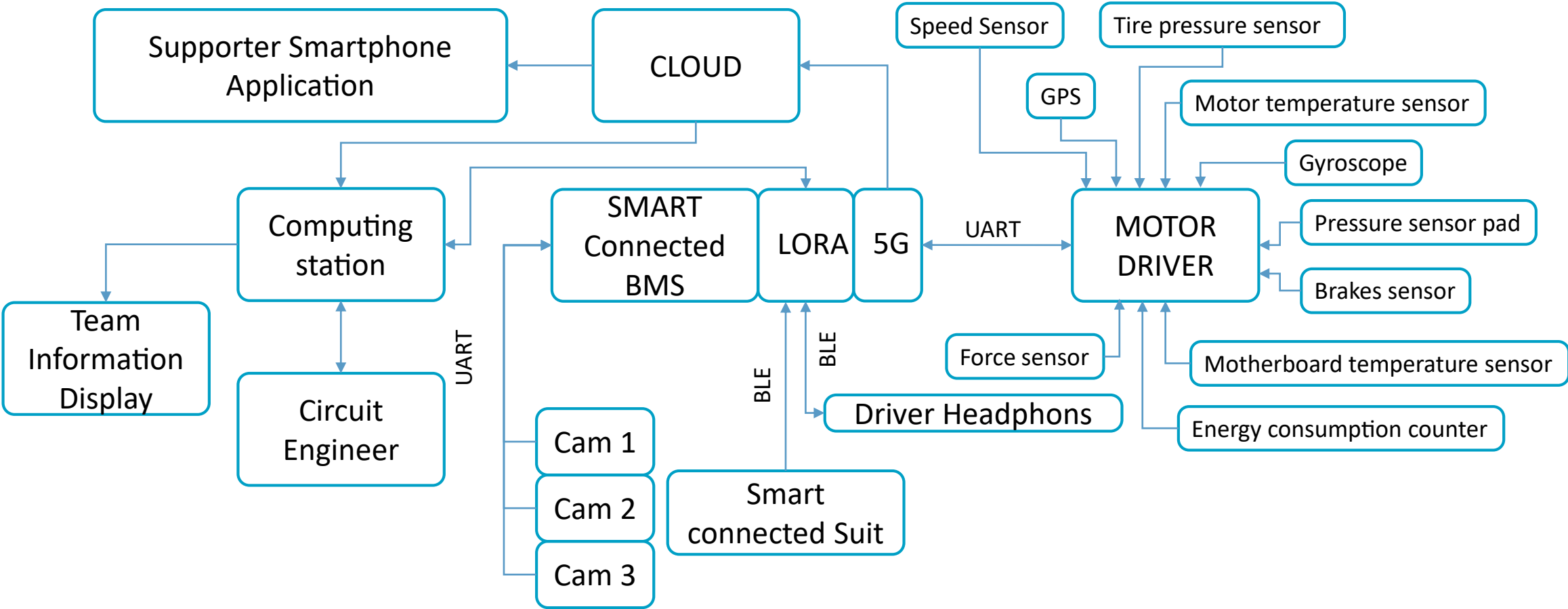
During the race all the data is recorded in the server and cloud in the form of log files that will be later processed to prepare the strategy for the next race attempts based on indicators and statistics and to map the track and create a faithful and realistic environment model to be used to improve the systems and simulate them in realistic conditions during the next editions of competition.



C o m m u n i c a t e d D A T A

All the data from the sensors installed in the vehicle are centralized at the vehicle controller and transmitted via a series connection to the Smart Connected BMS, which in addition to its main function of controlling the battery, acts as a communication interface between the driver and the team. It transmits the battery data and other data provided by all the components to the calculation server and the cloud, and receives the instructions to be displayed on a HUD Display for the driver via a 5G or Loara Connection. The data received from the cloud and from the calculation server will be pro-

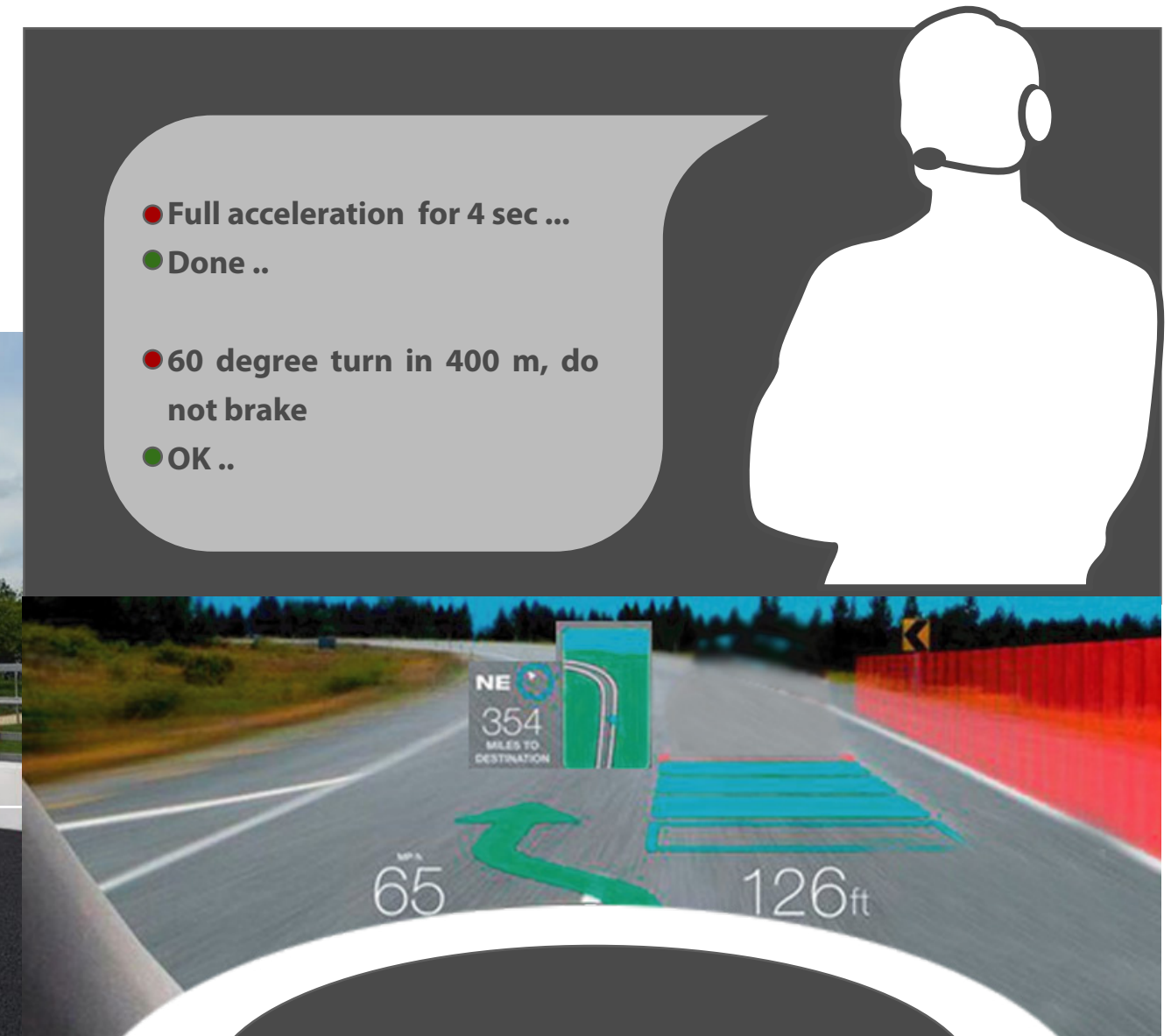
cessed and displayed as graphs and indicators in Full mode for the team on their smartphones via a dedicated application. On the other hand, the server sends autonomously to the driver's HUD Display the instructions in soft mode generated in real time by an algorithm on the basis of the data received from the vehicle. If necessary or it's an emergency , the circuit engineer from the communication station can make decisions according to the indicators and communicate them to the driver via audio or via instructions displayed on his HUD Display.



C o m m u n i c a t e d D A T A

What will the driver do with the data?

Since the pilot manages the driving with all the normal actions to be taken on the track, he tries to follow the maximum of the instructions generated and sent by the calculation server on his HUD screen as well as the one communicated by the circuit engineer via Audio or displayed on the HUD screen to guarantee a better consumption management during the race. It can also signal an accident or a problem of a parameter not measured by the sensors installed in the prototype.



HUD Display

Expected outcome from the use of the data

This flow of collected data in real-time will help the driver take the right decisions when handling the vehicle in order to conserve energy, and it will help the team to predict if a problem may occur at the time of the race, they can anticipate and avoid a possible risk threatening the driver or the vehicle's equipment. They can also help to create a database for testing and simulations to be put in place for future races.

The main purpose of using the data is to minimize the energy consumed during acceleration by avoiding overloading the motor during the passage through the upward slopes of the track and by preventing more speed loss during braking.

Today, with artificial intelligence and algorithms for calculating the ideal speed in comparison with the map drawn during the previous lap, the vehicle can cover the maximum possible distance with its own inertia without having to touch the brakes, so the driver only accelerates in very precise positions so as to not consume more energy in unnecessary use.

